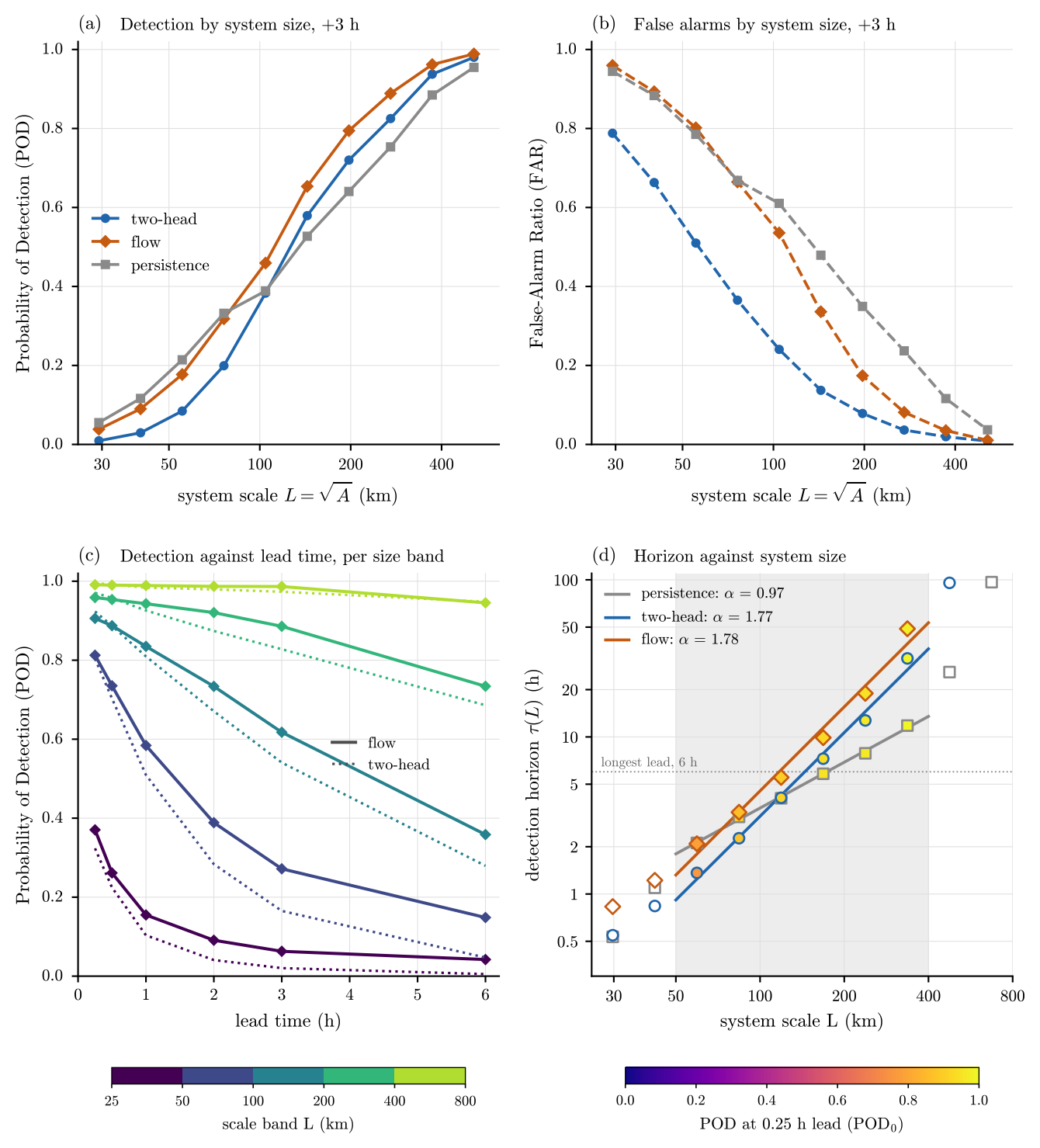
(a) Detection by system size, +3 h
Probability of Detection (POD)
system scale $L = \sqrt{A}$ (km)
two-head
flow
persistence
(b) False alarms by system size, +3 h
False-Alarm Ratio (FAR)
system scale $L = \sqrt{A}$ (km)
(c) Detection against lead time, per size band
Probability of Detection (POD)
lead time (h)
flow
two-head
scale band L (km)
(d) Horizon against system size
detection horizon $\tau(L)$ (h)
persistence: $\alpha$ = 0.97
two-head: $\alpha$ = 1.77
flow: $\alpha$ = 1.78
longest lead, 6 h
system scale L (km)
POD at 0.25 h lead ($POD_0$)